# Standard chemoradiotherapy with concurrent and adjuvant camrelizumab in patients with high risk nasopharyngeal carcinoma: multicentre, randomised, open label, phase 3 trial

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## ABSTRACT

### OBJECTIVE
To assess treatment with camrelizumab (a programmed death 1 inhibitor) in addition to concurrent chemoradiotherapy and as a maintenance treatment in patients with high risk nasopharyngeal carcinoma.

### DESIGN
Multicentre, randomised, open label, phase 3 trial.

### SETTING
Seven hospitals in China between 18 August 2020 and 21 June 2022.

### PARTICIPANTS
Adults aged 18-70 years with newly diagnosed high risk nasopharyngeal carcinoma after three cycles of induction chemotherapy with gemcitabine and cisplatin (stage 4a, stage 2-3 with stable or progressive disease, or detectable Epstein-Barr virus DNA).

### INTERVENTIONS
Patients were randomly assigned (1:1) to receive combination chemoradiotherapy based on cisplatin (standard treatment group) or standard treatment with 19 cycles of intravenous camrelizumab (200 mg) once every three weeks (radiotherapy plus two concurrent cycles and 17 adjuvant cycles; camrelizumab group).

### MAIN OUTCOME MEASURES
The primary endpoint in the intention-to-treat group was progression-free survival, defined as the time from randomisation to disease recurrence (locoregional or distant) or death from any cause. Secondary endpoints included safety and overall survival.

### RESULTS
390 patients were enrolled and randomly assigned to the camrelizumab group (n=194) or the standard treatment group (n=196). At median follow-up of 39.9 months (interquartile range 36.8-43.4 months), progression-free survival was higher in the camrelizumab group than the standard treatment group (36 months: 83.4%, 95% confidence interval 78.3% to 88.8% *v* 71.3%, 65.2% to 77.9%; stratified hazard ratio 0.51, 95% confidence interval 0.34 to 0.77, P=0.001). The incidence of acute and late adverse events (grade 3 or 4) was 50.5% and 3.2% in the camrelizumab group compared with 48.7% and 3.7% in the standard treatment group. Immunological adverse events (grade 3 or 4) occurred in 19 patients (10.2%) in the camrelizumab group.

### CONCLUSION
The addition of camrelizumab to concurrent chemoradiotherapy and as a maintenance treatment improved progression-free survival among patients with high risk nasopharyngeal carcinoma after induction chemotherapy.

### TRIAL REGISTRATION
ClinicalTrials.gov NCT04453826

## WHAT IS ALREADY KNOWN ON THIS TOPIC

Two previous phase 3 trials evaluated a combination of PD-1 (programmed death 1) blockade and induction chemotherapy with concurrent chemoradiotherapy in locoregionally advanced nasopharyngeal carcinoma

In one trial, a PD-1 inhibitor was used throughout the entire process, and in the second trial, PD-1 blockade was used as adjuvant treatment

## WHAT THIS STUDY ADDS

This phase 3, randomised clinical trial investigated the effect of PD-1 inhibitors in combination with concurrent chemoradiotherapy and as adjuvant treatment for high risk nasopharyngeal carcinoma

The addition of camrelizumab to concurrent chemoradiotherapy and as maintenance treatment was shown to improve progression-free survival in patients with high risk nasopharyngeal carcinoma after induction chemotherapy

This study validates the precision treatment concept of using dynamic imaging assessment and Epstein-Barr virus DNA changes to individualise subsequent treatment intensity in nasopharyngeal carcinoma

## Introduction

Nasopharyngeal carcinoma is a locally prevalent head and neck cancer, with a characteristic geographical distribution concentrated in South China, South East Asia, and North Africa. In 2022, nasopharyngeal carcinoma affected an estimated 120 416 patients worldwide.[1] Chemotherapy combined with radiotherapy is a crucial advance in the treatment of locoregionally intermediate and advanced disease. The National Comprehensive Cancer Network guidelines (version 1, 2020) recommend concurrent chemoradiotherapy with adjuvant chemotherapy or induction chemotherapy followed by concurrent chemoradiotherapy, which is supported by level 2A evidence for stage 2-4a disease.[2] Compared with sequencing of adjuvant chemotherapy, induction chemotherapy is better tolerated and further improves the control of distant disease for better survival, which plays an increasingly important role in managing nasopharyngeal carcinoma during intensity modulated radiotherapy.[3] However, recurrence is observed in 20-30% of patients despite this intensified treatment approach, highlighting the need for new clinical modalities tailored to this high risk population.[4-6]

Previous risk stratification tools largely focused on pretreatment factors such as staging based on imaging before treatment.[7] However, considerable heterogeneity remains for most cancer subtypes—patients with similar pretreatment staging may have different survival outcomes despite receiving the same treatment.[8] An early response to systemic treatment is a strong prognostic factor in many cancers, including haematological cancers and solid tumours.[9-12] Retrospective studies have indicated that the tumour response after induction chemotherapy based on radiological examination was an independent prognostic indicator of survival outcomes for patients with stage 2-4a nasopharyngeal carcinoma. The survival rates of patients who achieved complete or partial response (defined as radiological response) after induction chemotherapy were shown to be much higher than in patients evaluated to have stable disease or disease progression (defined as no radiological response).[13] In addition to radiological evaluation, the clearance status of Epstein-Barr virus DNA (EBV DNA) after induction chemotherapy also holds significance. Patients with detectable EBV DNA copy numbers after induction chemotherapy (defined as no biological response) had poorer survival outcomes than those with undetectable EBV DNA (defined as biological response).[14 15] Additionally, patients with stage T4 or N3 cancer intrinsically have the worst survival outcomes, except for those staged as M1.[16 17] Therefore, patients with stage 2-3 disease without radiological or biological response after induction chemotherapy or with stage T4 or N3 disease regardless of the response status after induction chemotherapy can potentially be considered to be at high risk.

In search of new treatments, several phase 3 trials have shown that adding programmed death 1 (PD-1) inhibitors to chemotherapy as first line treatment for recurrent or metastatic nasopharyngeal carcinoma provided statistically significantly and clinically meaningful benefits in terms of progression-free and overall survival compared with chemotherapy alone.[18-20] Therefore, the addition of PD-1 inhibitors to the chemoradiotherapy regimen after induction chemotherapy may be a promising treatment strategy for this high risk population. Recently, the CONTINUUM study confirmed our hypothesis that the addition of the PD-1 inhibitor sintilimab to chemoradiotherapy improved progression-free survival with manageable adverse events.[21]

Camrelizumab is a fully humanised, high affinity monoclonal antibody that binds to PD-1, and has shown efficacy in nasopharyngeal carcinoma.[22] We assessed the efficacy and safety of adding camrelizumab to chemoradiotherapy after induction chemotherapy in patients with high risk nasopharyngeal carcinoma evaluated as having no radiological or no biological response, and those with cancer staged as T4 or N3 regardless of response.

## Methods

### Study design

This multicentre, randomised, open label, phase 3 trial was conducted in seven hospitals in China (supplementary table 1). The study protocol was approved by the ethics committees of each participating centre. The trial conformed to the Declaration of Helsinki and the reporting of results was based on the CONSORT (consolidated standards of reporting trials) statement. The trial was registered with ClinicalTrials.gov (NCT04453826).

### Participants

Eligible patients were aged 18-70 years with histologically confirmed non-keratinising nasopharyngeal carcinoma (differentiated or undifferentiated type). Other inclusion criteria were stage 4a (T4 or N3) at diagnosis (according to the staging system of the American Joint Committee on Cancer, eight edition); stage 2-3 evaluated as stable disease or progressive disease (no radiological response) or detectable EBV DNA (no biological response) after induction chemotherapy; Karnofsky performance status score of 70 or higher; and adequate organ function. Exclusion criteria included recurrent or metastatic nasopharyngeal carcinoma; any active autoimmune disease or history of autoimmune disease; systemic immunosuppressive therapy; positive for hepatitis B surface antigen with more than 1000 copies of hepatitis B virus DNA per millilitre; and infection with HIV or hepatitis C. Eligibility assessment and obtaining written informed consent from eligible patients was the responsibility of investigators at each centre.

### Randomisation and masking

Patients were enrolled after completing three cycles of induction chemotherapy, stratified by treatment centre and disease stage (2-3 *v* 4a), and were randomly assigned (1:1) to receive cisplatin based concurrent chemoradiotherapy (standard treatment group) or concurrent chemoradiotherapy plus concurrent and adjuvant camrelizumab (camrelizumab group). The centralised randomisation procedure was conducted in the current study. Only the study coordinator and statistician knew the block structure, and they had no involvement in clinical aspects of the trial. The treatment group assignment was known to the investigators and patients (open label), but this was masked to the central imaging review committee members (W-JZ, L-ZL), who reviewed the imaging results of patients without access to information about treatment.

### Procedures

All patients were administered three cycles of induction chemotherapy comprising intravenous cisplatin (80 mg/m$^2$ on day 1) and gemcitabine (1 g/m$^2$ on days 1 and 8 of each 21 day cycle) once every three weeks. After induction chemotherapy, patients received two cycles of intravenous cisplatin (total of 200 mg/m$^2$) on days 1 and 22 of intensity modulated radiotherapy. Because previous studies indicated that patients who received concurrent cisplatin (200 mg/m$^2$) in two cycles experienced reduced toxic side effects while maintaining similar survival compared with those treated with three cycles,[23-26] the decision was made to use two cycles. No administration of off-protocol anticancer drugs was allowed before protocol defined disease progression. The supplementary appendix lists details of the chemotherapy dose, any modifications to the dose, and supportive measures.

Both groups received mandatory intensity modulated radiotherapy based on guidelines provided in the supplementary appendix. The recommendation was to start chemoradiotherapy within 21-28 days after the first day of the last cycle of induction chemotherapy.

Patients in the camrelizumab group received 200 mg camrelizumab intravenously over a minimum duration of 60 minutes on the initial day of radiotherapy, once every three weeks for two cycles during intensity modulated radiotherapy. These patients continued to receive maintenance treatment up to one year after the completion of radiotherapy (comprising 17 adjuvant cycles), unacceptable toxicity, start of new anticancer treatment, radiographic progression, withdrawal of consent, or investigator decision, whichever occurred first. The camrelizumab dose was fixed and no modification was permitted. When a grade 3 adverse event occurred and was associated with camrelizumab, as determined by the investigator, treatment was delayed until the adverse event was reduced to grade 1 or 0; camrelizumab treatment was permanently discontinued if a grade 4 adverse event occurred. If chemotherapy was discontinued because of adverse events unrelated to camrelizumab, camrelizumab treatment could be continued, while chemotherapy could be continued in patients who discontinued camrelizumab owing to adverse events unrelated to chemotherapy.

Pretreatment evaluation included a complete history and physical examination, haematological and biochemical assessments, fibreoptic nasopharyngoscopy, and magnetic resonance imaging of the nasopharynx and neck. Exclusion of distant metastasis was based on 18F fluorodeoxyglucose positron emission tomography computed tomography or conventional examinations including computed tomography of the chest and abdomen, combined with a bone scan.

The evaluation of radiological response relied on flexible nasopharyngoscopy and magnetic resonance imaging of the nasopharynx and neck, one week after completion of induction chemotherapy and 16 weeks after chemoradiotherapy, and was based on the RECIST criteria (response evaluation criteria in solid tumours; version 1.1). The radiological response of all recruited patients was independently evaluated by the central imaging review committee, who had no knowledge of group assignment. The biological response assessment was based on the EBV DNA testing before treatment and one week after the completion of induction chemotherapy. All blood samples were transported in a cold chain from the subcentres to Sun Yat-sen University Cancer Center for centralised detection of EBV DNA (supplementary appendix). Those evaluated as having stable or progressive disease (no radiological response) or detectable EBV DNA (no biological response) after induction chemotherapy were considered as non-responders to induction chemotherapy while those evaluated as having partial or complete response (radiological response) and undetectable EBV DNA (biological response) after induction chemotherapy were defined as responders. The evaluation method of response was the same for patients with stage 2-3 and stage 4a disease.

Follow-ups were conducted every three months for the first three years, after which the period was prolonged to six months. The attending physician assessed and confirmed all clinical outcomes. When feasible, fine needle aspiration or biopsy of suspected lesions was performed to confirm distant or locoregional disease progression. When tumour progression was confirmed, salvage treatments (eg, surgery or chemotherapy) were administered when possible.

Every visit included safety evaluations based on adverse events reported by patients, laboratory examination, physical examination, magnetic resonance imaging, and fibreoptic nasopharyngoscopy. The grading of acute toxicities was based on the Common Terminology Criteria for Adverse Events (CTCAE, version 5.0), while radiation related late toxicities (starting at three months after completion of radiotherapy) were graded based on a combination of the Radiation Therapy Oncology Group criteria and the CTCAE (version 5.0). Events occurring within 100 days after the last administration of camrelizumab, without clear causes but with potential immune mediated mechanisms or treated with immunomodulatory drugs (or hormone therapy), were classified as adverse events related to camrelizumab.

Quality of life assessment data were collected before starting chemoradiotherapy, during radiotherapy, at the end of radiotherapy, and one year after the end of radiotherapy using the Quality of Life Core questionnaire with 30 items developed by the European Organisation for Research and Treatment of Cancer (EORTC QLQ-C30).

### Outcomes

Progression-free survival was the primary endpoint, defined as the time from randomisation to the first documented evidence of disease recurrence (locoregional disease recurrence or distant metastasis) or death from any cause. Secondary endpoints included overall survival, distant metastasis-free survival, locoregional relapse-free survival, treatment response, treatment adherence, safety, and quality of life. Imaging results for assessing the secondary endpoints of locoregional relapse-free survival and distant metastasis-free survival were centrally reviewed. Locoregional relapse-free survival was defined as the time from randomisation to locoregional relapse or death from any cause. Distant metastasis-free survival was defined as the time from randomisation to distant metastasis or death from any cause. Patients with simultaneous distant and locoregional disease progression were included in both statistical groups. Data of patients who remained alive without recorded events were subjected to right censoring at the last follow-up date, with 15 February 2025 as the data cut off.

Responses to items on the EORTC QLQ-C30 questionnaire[27] were scored using scales representing symptoms or functions, with mean point transfer of all items pertaining to a domain, followed by transformation to a 0-100 scale based on the instructions of the EORTC scoring manual. Higher scores on the global health status and functioning scales were indicative of better health and function, while higher scores on the symptom scales indicated worse symptoms.

Prespecified exploratory endpoints included progression-free survival, locoregional relapse-free survival, and distant metastasis-free survival in subgroups defined by pretreatment characteristics and clinical or biological markers.

### Statistical analysis

The primary aim of this study was to test whether chemoradiotherapy plus camrelizumab improved progression-free survival over chemoradiotherapy alone. Based on previous studies, we assumed that the three year progression-free survival rate would be 75% for patients treated with chemoradiotherapy alone,[4] and 85% for patients receiving camrelizumab in addition to chemoradiotherapy, corresponding to a targeted hazard ratio of 0.565. The sample size was calculated using the log rank test. The expected length of the accrual period and maximal length of follow-up were three and six years, respectively. The power was 0.80 ($1-\beta$) and the two sided type I error ($\alpha$) was 0.05. Based on a presumed 10% dropout rate, a total of 388 participants were estimated to be required (194 participants in each group), allowing for 98 events in the primary analysis of progression-free survival. The study mandated interim analyses by an independent data monitoring committee. In the interim analyses based on 49 observed events, type 1 errors were strictly controlled using the O'Brien-Fleming spending function. This interim analysis used a two sided significance level of 0.003. The committee reviewed the analyses and approved the continuation of the trial.

The primary efficacy assessment included all patients who were randomly assigned to a group (the intention-to-treat population). The reverse Kaplan-Meier method was used to calculate the median follow-up time. Time-to-event data were visualised using Kaplan-Meier curves, and compared using the stratified log rank test. The hazard ratio and 95% confidence intervals (CIs) were calculated using a Cox proportional hazards model, with covariates including the overall stage (2-3 *v* 4), treatment group, and trial centre, while the proportional hazards assumption was tested using Schoenfeld residuals.[28] We further performed an interaction analysis to explore whether the effect of the experimental treatment varied in the subgroups defined according to sex, age, body mass index, smoking status, drinking status, overall stage, T stage, N stage, baseline EBV DNA copy number, EBV DNA clearance status after induction chemotherapy, radiological evaluation after induction chemotherapy, overall response, stage, and response. Among these factors, age, body mass index, and baseline EBV DNA copy number were dichotomised by the median. The interaction analysis was conducted by means of a test of treatment-by-covariate interaction on the Cox proportional hazards model.

The safety population comprised patients who received actual protocol conformant treatment and formed the basis for safety analysis. We used frequencies and percentages to summarise the rates of severe late complications. The health related quality of life data were analysed using a mixed effects model, with response values consisting of the scores assessed at each visit, as well as fixed effects encompassing the baseline score, treatment group, visit, and stratification factors related to randomisation (stage and centre) without any interaction items, while the subject was treated as a random effect. The treatment coefficient was calculated as the difference in average baseline adjusted quality of life scores between groups. The interaction effects of the visit and treatment group were evaluated by adding their interaction item to the model. The average effect was assessed based on the average differences of all visit times, and the difference at each visit was visualised using profile plots.

Analyses were performed using SPSS software (version 25.0; IBM, Armonk, NY, USA) and R version 4.4.1. The threshold for statistical significance was a two tailed P value less than 0.05. The secondary endpoints were not subjected to multiple adjustments and were only used for exploratory analysis.

### Patient and public involvement

We are aware that patient and public involvement would be a crucial contribution to our research, but there was no funding for this activity. Nevertheless, the needs and preferences of the patients were considered during planning, and results will be shared through social media and explained to participants. All patients were aware of the trial objectives and contents on recruitment. Additionally, we are also considering incorporating structured patient and public involvement into our trials in the future.

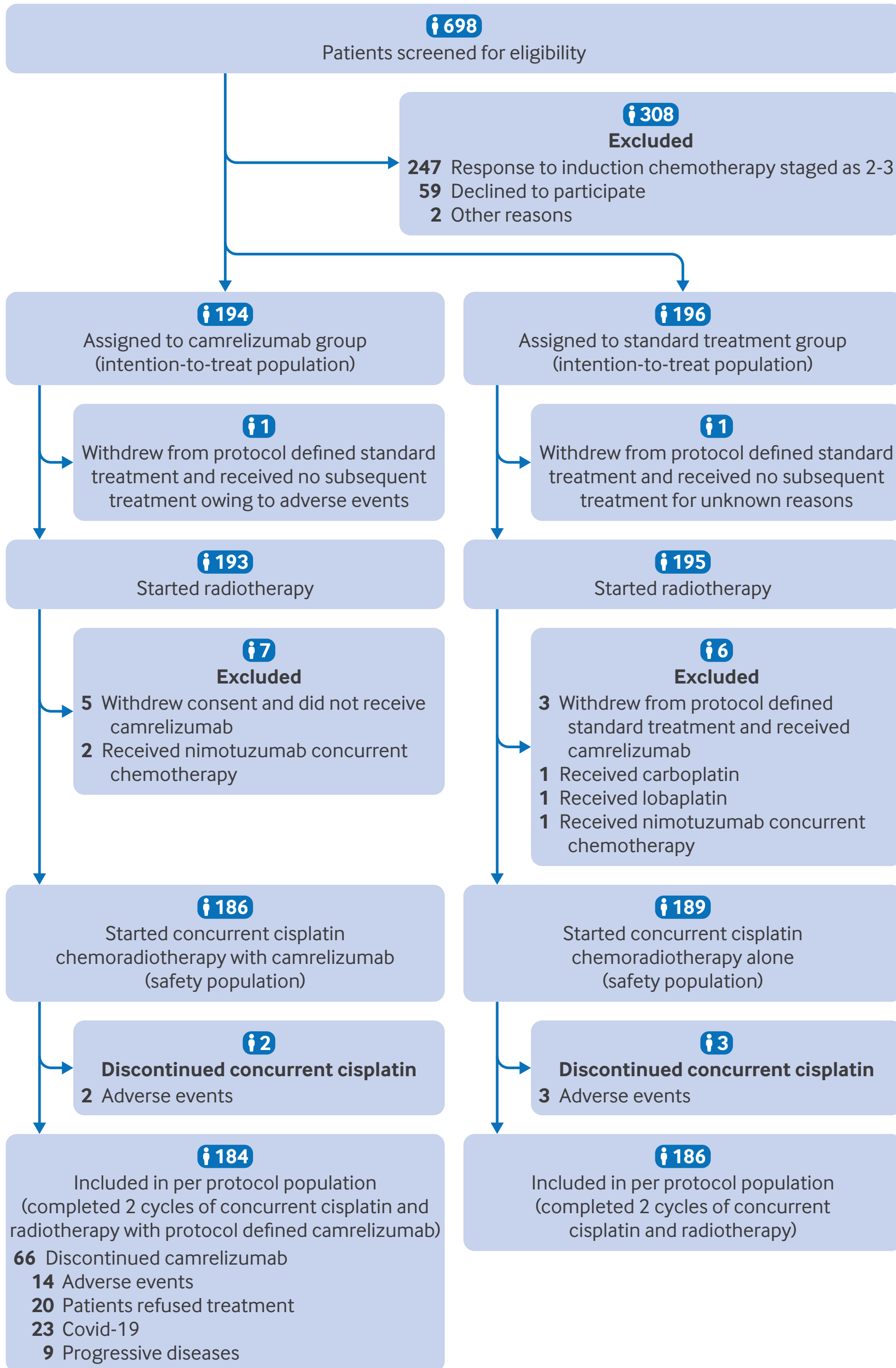


**Fig 1 | Flow chart showing patients and treatment groups**

## Results

### Patients and treatment

Between 18 August 2020 and 21 June 2022, 698 patients were screened for eligibility, 390 of whom were enrolled and randomly assigned to the standard treatment group (n=196) or the camrelizumab group (n=194; intention-to-treat population; fig 1). In the standard treatment group, 54 patients had stage 2-3 disease, all of whom showed no radiological or biological response to induction chemotherapy (radiological evaluation as stable or progressive disease or detectable EBV DNA). Among the 142 patients with stage 4a disease, 51 showed no radiological or biological response to induction chemotherapy, while 91 showed a response (radiological evaluation as complete or partial response and undetectable EBV DNA). In the camrelizumab group, 52 patients had stage 2-3 disease with no radiological or biological response to induction chemotherapy. Of the 142 patients with stage 4a disease, 48 showed no radiological or biological response to induction chemotherapy, while 94 showed a response.

Table 1 | Baseline characteristics of patients

| Characteristics | Camrelizumab group (n=194) | Standard treatment group (n=196) |
|---|---|---|
| Sex | | |
| Female | 57 (29.4) | 50 (25.5) |
| Male | 137 (70.6) | 146 (74.5) |
| Age (years), median (IQR) | 46.0 (36.8-54.0) | 45.0 (35.0-54.0) |
| Karnofsky performance status score | | |
| 90-100 | 192 (99.0) | 193 (98.5) |
| 70-80 | 2 (1.0) | 3 (1.5) |
| Body mass index, median (IQR) | 21.7 (19.6-24.6) | 22.9 (20.7-25.5) |
| Histology | | |
| Non-keratinising undifferentiated | 193 (99.5) | 193 (98.5) |
| Non-keratinising differentiated | 1 (0.5) | 3 (1.5) |
| EBV DNA copy number at diagnosis, median (IQR) | 1237.0 (249.3-6730.0) | 1130.0 (178.8-7375.0) |
| T classification | | |
| T1 | 6 (3.1) | 10 (5.1) |
| T2 | 17 (8.8) | 15 (7.7) |
| T3 | 94 (48.5) | 87 (44.4) |
| T4 | 77 (39.7) | 84 (42.9) |
| N classification | | |
| N0 | 7 (3.6) | 6 (3.1) |
| N1 | 46 (23.7) | 46 (23.5) |
| N2 | 61 (31.4) | 70 (35.7) |
| N3 | 80 (41.2) | 74 (37.8) |
| TNM stage | | |
| 2 | 7 (3.6) | 9 (4.6) |
| 3 | 45 (23.2) | 45 (23.0) |
| 4a | 142 (73.2) | 142 (72.4) |
| Smoking (current or former) | | |
| Yes | 56 (28.9) | 62 (31.6) |
| No | 138 (71.1) | 134 (68.4) |
| Drinking (current or former) | | |
| Yes | 25 (12.9) | 28 (14.3) |
| No | 169 (87.1) | 168 (85.7) |
| Response to induction chemotherapy | | |
| No response | 100 (51.5) | 105 (53.6) |
| Response | 94 (48.5) | 91 (46.4) |
| Family history of nasopharyngeal carcinoma | | |
| Yes | 5 (2.6) | 6 (3.1) |
| No | 189 (97.4) | 190 (96.9) |

Data are numbers (%) unless stated otherwise.
EBV=Epstein-Barr virus; IQR=interquartile range.

All baseline characteristics were similar in both groups. The median age was 46.0 (interquartile range 36.8-54.0) in the camrelizumab group and 45.0 (35.0-54.0) in the standard treatment group. The population consisted of 107 women (27.4%) and 283 men (72.6%). Additionally, 77 of 194 patients (39.7%) had T4 disease in the camrelizumab group compared with 84 of 196 (42.9%) in the standard treatment group. Eighty of 194 patients (41.2%) had N3 disease in the camrelizumab group compared with 74 of 196 (37.8%) in the standard treatment group (table 1 and supplementary table 2).

Among 194 patients assigned to the camrelizumab group, one withdrew from the protocol defined standard treatment and received no subsequent treatment owing to adverse events, five did not receive camrelizumab, while two received concurrent nimotuzumab chemotherapy. Of the 196 patients assigned to the standard treatment group, one withdrew from the protocol defined standard treatment and received no subsequent treatment for unknown reasons, three received camrelizumab, three did not receive concurrent cisplatin (one received carboplatin and one lobaplatin, while one received concurrent nimotuzumab chemotherapy). Overall, 186 patients (95.9%) in the camrelizumab group and 189 (96.4%) in the standard treatment group started protocol defined treatment (safety population; fig 1). In the camrelizumab group, during the concurrent phase, 184 patients (94.8%) received two cycles of concurrent camrelizumab treatment. During the adjuvant phase, 120 patients (61.9%) completed the protocol defined 17 cycles of camrelizumab maintenance treatment. The median number of camrelizumab cycles in the adjuvant phase was 17 (interquartile range 9-17), 27 patients (13.9%) received three or fewer cycles, 15 patients (7.7%) received four to six cycles, nine patients (4.6%) received seven to nine cycles, and eight patients (4.1%) received 10-16 cycles. Reasons for camrelizumab discontinuation included patient refusal (20, 10.3%), adverse events (14, 7.2%), insufficient medical resources during the covid-19 pandemic (23, 11.9%), and disease progression (9, 4.6%; supplementary table 3; fig 1).

Table 2 | Survival and response to treatment

| Survival and response | Camrelizumab group (n=194) | Standard treatment group (n=196) | Difference in rates at 36 months, % (95% CI) | Hazard ratio* (95% CI) | Log rank P value |
|---|---|---|---|---|---|
| **Progression-free survival** | | | | | |
| Events | 36 (18.6) | 64 (32.7) | — | — | — |
| Rate at 36 months (95% CI) | 83.4 (78.3 to 88.8) | 71.3 (65.2 to 77.9) | 12.1 (3.9 to 20.4) | 0.51 (0.34 to 0.77) | 0.001 |
| **Overall survival** | | | | | |
| Events | 10 (5.2) | 17 (8.7) | — | — | — |
| Rate at 36 months (95% CI) | 94.8 (91.7 to 98.0) | 92.2 (88.5 to 96.1) | 2.6 (−2.3 to 7.5) | 0.59 (0.27 to 1.28) | 0.19 |
| **Distant metastasis-free survival** | | | | | |
| Events | 23 (11.9) | 39 (19.9) | — | — | — |
| Rate at 36 months (95% CI) | 88.9 (84.5 to 93.5) | 80.7 (75.2 to 86.6) | 8.2 (1.0 to 15.5) | 0.54 (0.32 to 0.90) | 0.02 |
| **Locoregional recurrence-free survival** | | | | | |
| Events | 15 (7.7) | 35 (17.9) | — | — | — |
| Rate at 36 months (95% CI) | 92.8 (89.1 to 96.6) | 83.2 (77.9 to 88.9) | 9.6 (2.9 to 16.3) | 0.38 (0.21 to 0.70) | 0.001 |
| **Response to induction therapy** | | | | | |
| Complete response | 1 (0.5) | 0 (0.0) | — | | |
| Partial response | 138 (71.1) | 134 (68.4) | | | |
| Stable disease | 55 (28.4) | 62 (31.6) | | | |
| Progressive disease | 0 (0.0) | 0 (0.0) | | | |
| **Response to total treatment** | | | | | |
| Complete response | 175 (90.2) | 177 (90.3) | — | | |
| Partial response | 10 (5.2) | 12 (6.1) | | | |
| Stable disease | 1 (0.5) | 1 (0.5) | | | |
| Progressive disease | 3 (1.5) | 3 (1.5) | | | |
| Could not be assessed | 5 (2.6) | 3 (1.5) | | | |

Data are numbers (%) unless stated otherwise. CI=confidence interval.
*Hazard ratios were calculated using a stratified proportional hazards model.

Three cycles of gemcitabine and cisplatin induction chemotherapy were completed in 194 patients (100.0%) in the camrelizumab group and 196 (100.0%) in the standard treatment group. Of these, 42 of 194 patients (21.6%) received dose reduction of gemcitabine and cisplatin induction chemotherapy in the camrelizumab group, which was comparable to 40 of 196 (20.4%) in the standard treatment group (supplementary table 4). In the concurrent chemoradiotherapy phase, 191 of 194 patients (98.5%) in the camrelizumab group completed two cycles of concurrent cisplatin, which was similar to 192 of 196 patients (98.0%) in the standard treatment group. The cumulative concurrent cisplatin dose was 188.9 mg/m$^2$ (interquartile range 169.5-200.0) in the camrelizumab group and 189.6 mg/m$^2$ (161.3-200.0) in the standard treatment group (supplementary table 5).

In terms of radiotherapy, 193 of 194 patients (99.5%) in the camrelizumab group and 195 of 196 (99.5%) in the standard treatment group completed protocol defined intensity modulated radiotherapy. The median time from the start of the last induction chemotherapy cycle to the start of chemoradiotherapy was 24 days (interquartile range 22-30) in the camrelizumab group and 24 days (22-31) in the standard treatment group (supplementary table 4). The prescribed dose, duration of radiotherapy, and dosimetric parameters were similar in the two groups (supplementary tables 6 and 7).

## Efficacy

Overall, 71.1% (138 of 194) of patients in the camrelizumab group and 68.4% (134 of 196) in the standard treatment group had a partial response, while 28.4% (55 of 194) of patients in the camrelizumab group and 31.6% (62 of 196) in the standard treatment group had stable disease after three cycles of induction chemotherapy. At 16 weeks after radiotherapy, 175 of 194 patients (90.2%) in the camrelizumab group and 177 of 196 (90.3%) in the standard treatment group had a complete response (table 2).

The final follow-up was on 15 February 2025, corresponding to a median follow-up time of 39.9 months (interquartile range 36.8-43.4). As of this date, 318 of 363 patients were alive (87.6%) with follow-up records of at least 36 months, while the last enrolled patient had a final follow-up time of 29.4 months. A total of 100 patients experienced recurrence or died (25.6% of the overall trial population), including 64 of 196 patients (32.7%) in the standard treatment group and 36 of 194 (18.6%) in the camrelizumab group (table 2). Supplementary tables 8 and 9 provide details of the relapse patterns and subsequent treatments. The 36 month progression-free survival rate was 83.4% (95% CI 78.3% to 88.8%) in the camrelizumab group compared with 71.3% (65.2% to 77.9%) in the standard treatment group (stratified hazard ratio for recurrence or death 0.51, 95% CI 0.34 to 0.77, P=0.001; table 2, fig 2).

At the time of analysis, 10 of 194 patients (5.2%) in the camrelizumab group and 17 of 196 patients (8.7%) in the standard treatment group had died. Supplementary table 8 provides details about the cause of death. No statistical difference in overall survival was found between the camrelizumab and standard treatment groups (94.8%, 95% CI 91.7% to 98.0% *v* 92.2%, 88.5% to 96.1%; stratified hazard

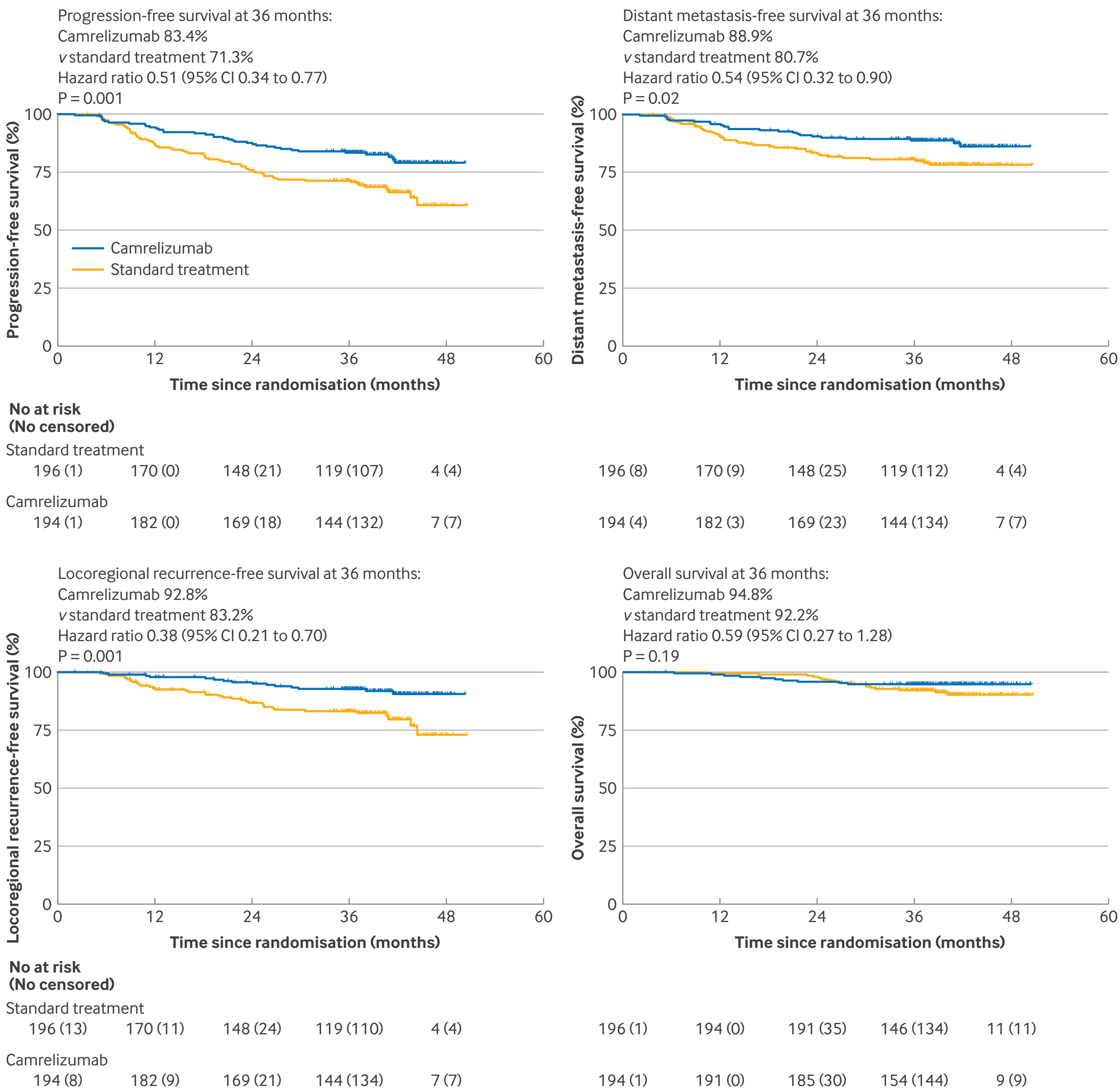


**Fig 2 | Kaplan-Meier survival analysis of intention-to-treat population at 36 months**

ratio for death 0.59, 95% CI 0.27 to 1.28; fig 2). We recorded a total of five deaths in the camrelizumab group between 12 and 24 months, including one from myocardial infarction, one from locoregional relapse, and three from multiple distant metastases. The 36 month distant metastasis-free survival rate was better in the camrelizumab group than in the standard treatment group (88.9%, 95% CI 84.5% to 93.5% *v* 80.7%, 75.2% to 86.6%; stratified hazard ratio for distant metastasis or death 0.54, 95% CI 0.32 to 0.90). Additionally, the 36 month locoregional recurrence-free survival rate was better in the camrelizumab group than in the standard treatment group (92.8%, 95% CI 89.1% to 96.6% *v* 83.2%, 77.9% to 88.9%; stratified hazard ratio for locoregional relapse or death 0.38, 95% CI 0.21 to 0.70; table 2, fig 2). Among patients who experienced locoregional relapse, 30 of 32 (93.8%) in the standard treatment group and 13 of 14 (92.9%) in the camrelizumab group had an in-field relapse (defined as 95% or more of the recurrence volume within the 95% isodose[29]).

### Adverse events

Throughout the entire treatment, acute adverse events of any grade were observed in all patients (100%) in the camrelizumab and standard treatment groups from the

**Table 3 | Adverse events and grade in two treatment groups**

| | Camrelizumab group (n=186) | | | | Standard treatment group (n=189) | | | |
|---|---|---|---|---|---|---|---|---|
| **Adverse events** | **Grade ≥3** | **Grade 1-2** | **Grade 3-4** | **Grade 5** | **Grade ≥3** | **Grade 1-2** | **Grade 3-4** | **Grade 5** |
| Any acute adverse events | 94 (50.5) | 92 (49.5) | 94 (50.5) | 0 (0.0) | 92 (48.7) | 97 (51.3) | 92 (48.7) | 0 (0.0) |
| Neutropenia | 28 (15.1) | 61 (32.8) | 28 (15.1) | 0 (0.0) | 28 (14.8) | 52 (27.5) | 28 (14.8) | 0 (0.0) |
| Leukopenia | 32 (17.2) | 78 (41.9) | 32 (17.2) | 0 (0.0) | 35 (18.5) | 78 (41.3) | 35 (18.5) | 0 (0.0) |
| Anaemia | 18 (9.7) | 161 (86.6) | 18 (9.7) | 0 (0.0) | 10 (5.3) | 165 (87.3) | 10 (5.3) | 0 (0.0) |
| Thrombocytopenia | 15 (8.1) | 81 (43.5) | 15 (8.1) | 0 (0.0) | 5 (2.6) | 80 (42.3) | 5 (2.6) | 0 (0.0) |
| Raised bilirubin | 0 (0.0) | 20 (10.8) | 0 (0.0) | 0 (0.0) | 0 (0.0) | 16 (8.5) | 0 (0.0) | 0 (0.0) |
| Raised ALT | 2 (1.1) | 44 (23.7) | 2 (1.1) | 0 (0.0) | 1 (0.5) | 37 (19.6) | 1 (0.5) | 0 (0.0) |
| Raised AST | 2 (1.1) | 30 (16.1) | 2 (1.1) | 0 (0.0) | 2 (1.1) | 23 (12.2) | 2 (1.1) | 0 (0.0) |
| Raised creatinine | 0 (0.0) | 56 (30.1) | 0 (0.0) | 0 (0.0) | 0 (0.0) | 46 (24.3) | 0 (0.0) | 0 (0.0) |
| Nausea | 9 (4.8) | 146 (78.5) | 9 (4.8) | 0 (0.0) | 8 (4.2) | 146 (77.2) | 8 (4.2) | 0 (0.0) |
| Vomiting | 9 (4.8) | 124 (66.7) | 9 (4.8) | 0 (0.0) | 9 (4.8) | 109 (57.7) | 9 (4.8) | 0 (0.0) |
| Diarrhoea | 1 (0.5) | 26 (14.0) | 1 (0.5) | 0 (0.0) | 1 (0.5) | 27 (14.3) | 1 (0.5) | 0 (0.0) |
| Fever | 0 (0.0) | 18 (9.7) | 0 (0.0) | 0 (0.0) | 0 (0.0) | 14 (7.4) | 0 (0.0) | 0 (0.0) |
| Dizziness | 0 (0.0) | 120 (64.5) | 0 (0.0) | 0 (0.0) | 0 (0.0) | 112 (59.3) | 0 (0.0) | 0 (0.0) |
| Fatigue | 1 (0.5) | 156 (83.9) | 1 (0.5) | 0 (0.0) | 2 (1.1) | 152 (80.4) | 2 (1.1) | 0 (0.0) |
| Dermatitis | 1 (0.5) | 152 (81.7) | 1 (0.5) | 0 (0.0) | 2 (1.1) | 160 (84.6) | 2 (1.1) | 0 (0.0) |
| Mucositis | 39 (21.0) | 118 (63.4) | 39 (21.0) | 0 (0.0) | 42 (22.2) | 139 (73.5) | 42 (22.2) | 0 (0.0) |
| Dry mouth | 0 (0.0) | 168 (90.1) | 0 (0.0) | 0 (0.0) | 0 (0.0) | 178 (94.2) | 0 (0.0) | 0 (0.0) |
| Inflammation of the throat | 0 (0.0) | 160 (86.0) | 0 (0.0) | 0 (0.0) | 1 (0.5) | 159 (84.1) | 1 (0.5) | 0 (0.0) |
| Any immune related adverse events | 19 (10.2) | 135 (72.6) | 19 (10.2) | 0 (0.0) | NA | NA | NA | NA |
| Raised amylase | 9 (4.8) | 19 (10.2) | 9 (4.8) | 0 (0.0) | NA | NA | NA | NA |
| Hypothyroidism | 1 (0.5) | 43 (23.1) | 1 (0.5) | 0 (0.0) | NA | NA | NA | NA |
| Hyperthyroidism | 3 (1.6) | 61 (32.8) | 3 (1.6) | 0 (0.0) | NA | NA | NA | NA |
| Rash | 3 (1.6) | 41 (22.0) | 3 (1.6) | 0 (0.0) | NA | NA | NA | NA |
| Cherry hemangioma | 0 (0.0) | 111 (59.7) | 0 (0.0) | 0 (0.0) | NA | NA | NA | NA |
| Pneumonia | 1 (0.5) | 8 (4.3) | 1 (0.5) | 0 (0.0) | NA | NA | NA | NA |
| Rheumatoid arthritis | 1 (0.5) | 7 (3.8) | 1 (0.5) | 0 (0.0) | NA | NA | NA | NA |
| Myositis | 2 (1.1) | 6 (3.2) | 2 (1.1) | 0 (0.0) | NA | NA | NA | NA |
| Any late adverse events | 6(3.2) | 169 (90.9) | 6 (3.2) | 0 (0.0) | 7 (3.7) | 174 (92.1) | 7 (3.7) | 0 (0.0) |
| Eye disorders | 0 (0.0) | 44 (23.7) | 0 (0.0) | 0 (0.0) | 0 (0.0) | 52 (27.5) | 0 (0.0) | 0 (0.0) |
| Hearing impairment | 5 (2.7) | 90 (48.4) | 5 (2.7) | 0 (0.0) | 7 (3.7) | 86 (45.5) | 7 (3.7) | 0 (0.0) |
| Trismus | 0 (0.0) | 32 (17.2) | 0 (0.0) | 0 (0.0) | 0 (0.0) | 38 (20.1) | 0 (0.0) | 0 (0.0) |
| Dry mouth | 1 (0.5) | 124 (66.7) | 1 (0.5) | 0 (0.0) | 0 (0.0) | 126 (66.7) | 0 (0.0) | 0 (0.0) |
| Skin reaction | 0 (0.0) | 75 (40.3) | 0 (0.0) | 0 (0.0) | 0 (0.0) | 86 (45.5) | 0 (0.0) | 0 (0.0) |
| Neck tissue damage | 0 (0.0) | 61 (32.8) | 0 (0.0) | 0 (0.0) | 0 (0.0) | 70 (37.0) | 0 (0.0) | 0 (0.0) |
| Nasopharyngeal mucosal necrosis | 1 (0.5) | 0 (0.0) | 1 (0.5) | 0 (0.0) | 1 (0.5) | 0 (0.0) | 1 (0.5) | 0 (0.0) |

Data are numbers (%). ALT=alanine aminotransferase; AST=aspartate aminotransferase; NA=not applicable.

safety population (186 and 189 patients, respectively), whereas acute adverse events (grade 3 or 4) occurred in 94 (50.5%) versus 92 (48.7%), respectively. These acute adverse events included mucositis (39, 21.0% *v* 42, 22.2% in the camrelizumab and standard treatment groups, respectively), leukopenia (32, 17.2% *v* 35, 18.5%), neutropenia (28, 15.1% *v* 28, 14.8%), anaemia (18, 9.7% *v* 10, 5.3%), thrombocytopenia (15, 8.1% *v* 5, 2.6%), nausea (9, 4.8% *v* 8, 4.2%), and vomiting (9, 4.8% *v* 9, 4.8%; table 3).

Most immune related adverse events were grade 1 or 2, with grade 3 or 4 events occurring in 19 patients (10.2%) from the camrelizumab group. The most common grade 3 or 4 immune related adverse events were raised amylase (9, 4.8%), rash (3, 1.6%), hyperthyroidism (3, 1.6%), myositis (2, 1.1%), pneumonia (1, 0.5%), and rheumatoid arthritis (1, 0.5%). Ten of 186 patients (5.4%) in the camrelizumab group required steroid treatment; none required additional immune suppressants.

A total of 90.9% (169 of 186) of patients in the camrelizumab group and 92.1% (174 of 189) in the standard treatment group experienced any late toxic effects of grade 1 or 2. Moreover, one or more late toxic effects of grade 3 or 4 were observed in 3.2% (6 of 186 patients) in the camrelizumab group and 3.7% (7 of 189 patients) in the standard treatment group. Overall, the distribution of late toxic effects was similar between the two groups (table 3).

### Quality of life

The quality of life population included 149 (76.0%) patients in the standard treatment group and 144 (74.2%) in the camrelizumab group. At all valid time points (before starting radiotherapy, during treatment, at the end, and 12 months after the end of treatment), the completion rate of the EORTC QLQ-C30 questionnaire was more than 50% (table 4). The baseline characteristics of patients analysed and their quality of life scores before radiotherapy did not differ significantly between the groups (supplementary tables 10 and 11). No significant difference was found in the general quality of life and symptom burden between the two groups (table 4). Supplementary figure 1 presents the changes in quality of life indices over time.

Table 4 | Mean differences between treatment groups in quality of life scores using European Organisation for Research and Treatment of Cancer Quality of Life Core 30 item questionnaire

| Item | Camrelizumab group (n=144) | Standard treatment group (n=149) | Mean difference* (95% CI) | P value |
|---|---|---|---|---|
| **General quality of life (the higher the better)** | | | | |
| Physical functioning | 81.2 (78.8 to 83.7) | 82.2 (79.8 to 84.5) | −0.9 (−3.9 to 2.0) | 0.53 |
| Role functioning | 79.4 (75.9 to 82.9) | 82.1 (78.8 to 85.4) | −2.7 (−6.9 to 1.5) | 0.21 |
| Cognitive functioning | 84.6 (82.0 to 87.3) | 86.2 (83.6 to 88.7) | −1.5 (−4.8 to 1.7) | 0.36 |
| Emotional functioning | 82.1 (79.2 to 85.0) | 83.4 (80.6 to 86.3) | −1.3 (−4.9 to 2.2) | 0.46 |
| Social functioning | 74.9 (71.5 to 78.3) | 77.9 (74.6 to 81.1) | −2.9 (−7.0 to 1.1) | 0.16 |
| Global health status | 57.0 (54.5 to 59.6) | 59.9 (57.4 to 62.3) | −2.8 (−5.9 to 0.2) | 0.07 |
| **Symptom burden (the lower the better)** | | | | |
| Pain | 18.3 (15.4 to 21.1) | 17.9 (15.2 to 20.6) | 0.4 (−3.0 to 3.8) | 0.83 |
| Fatigue | 32.0 (28.7 to 35.4) | 29.5 (26.3 to 32.7) | 2.5 (−1.5 to 6.5) | 0.22 |
| Dyspnoea | 16.7 (13.6 to 19.9) | 15.3 (12.3 to 18.3) | 1.4 (−2.4 to 5.3) | 0.46 |
| Insomnia | 24.2 (20.6 to 27.7) | 23.3 (19.9 to 26.7) | 0.8 (−3.4 to 5.1) | 0.70 |
| Appetite loss | 34.8 (30.8 to 38.8) | 36.5 (32.7 to 40.3) | −1.7 (−6.5 to 3.1) | 0.49 |
| Nausea and vomiting | 21.7 (18.7 to 24.7) | 22.9 (20.0 to 25.7) | −1.2 (−4.8 to 2.5) | 0.53 |
| Constipation | 18.2 (15.1 to 21.2) | 18.8 (15.9 to 21.7) | −0.6 (−4.3 to 3.0) | 0.74 |
| Diarrhoea | 2.3 (0.9 to 3.8) | 3.3 (1.9 to 4.7) | −1.0 (−2.7 to 0.8) | 0.27 |
| Financial difficulties | 28.9 (25.2 to 32.6) | 25.3 (21.7 to 28.8) | 3.6 (−0.8 to 8.1) | 0.11 |

Data are scores (95% confidence interval). CI=confidence interval.
*Camrelizumab group minus standard treatment group; a negative mean difference indicates symptoms were worse in standard treatment group, while general quality of life was worse in camrelizumab group.

### Prespecified exploratory analyses

The per protocol population comprised patients who completed two cycles of concurrent cisplatin and radiotherapy with protocol defined camrelizumab in the camrelizumab group, and those who completed two cycles of concurrent cisplatin and radiotherapy in the standard treatment group. The baseline characteristics of the patients were well balanced (supplementary table 12). In the per protocol population, patients in the camrelizumab group also had better three year progression-free survival than those in the standard treatment group (83.6%, 95% CI 78.4% to 89.1% *v* 71.5%, 65.3% to 78.3%; stratified hazard ratio for recurrence or death 0.51, 95% CI 0.33 to 0.77, P=0.001). Similarly, the three year distant metastasis-free survival and locoregional relapse-free survival rates were better in the camrelizumab group than in the standard treatment group (stratified hazard ratio for distant metastasis or death 0.54, 95% CI 0.32 to 0.92; stratified hazard ratio for locoregional relapse or death 0.34, 0.18 to 0.65; supplementary figure 2).

Subgroup analysis was prespecified and conducted using the baseline characteristics, revealing no significant interactions related to progression-free survival or distant metastasis-free survival between the treatment groups and subgroups except for body mass index. Among the subgroups with body mass index <22.4, the progression-free survival and distant metastasis-free survival rates were both significantly superior in the camrelizumab group compared with the standard treatment group (supplementary figures 3-5). We also found that the progression-free survival of patients with stage 4a disease evaluated to respond to induction chemotherapy was superior to that of patients with stage 4a disease evaluated not to respond and even better than that of patients with stage 2-3 disease evaluated not to respond (supplementary figure 6 and supplementary table 13). Additionally, we conducted a sensitivity analysis excluding patients from these four small centres and obtained similar results (supplementary figure 7).

## Discussion

### Principal findings

In this phase 3, multicentre clinical study, the addition of camrelizumab to concurrent chemoradiotherapy and as maintenance treatment significantly improved the progression-free survival rate of patients with high risk nasopharyngeal carcinoma after induction chemotherapy. The progression-free survival benefit of camrelizumab was accompanied by a significantly decreased risk of locoregional recurrence and distant metastasis, although there were no statistically significant differences in overall survival at the data cut-off time point.

### Comparison with other studies

Our findings were similar to two previous phase 3 trials (CONTINUUM[21] and DIPPER[30]), which both confirmed the value of PD-1 blockade in treating nasopharyngeal carcinoma. However, several distinguishing features were found in the current phase 3 study. We used central imaging evaluation and central serial EBV DNA testing after induction chemotherapy to accurately screen patients for eligibility for the addition of PD-1 inhibitors, rather than relying solely on baseline radiological staging. Previous studies have shown that the radiological response and EBV DNA clearance status after induction chemotherapy could effectively be used to screen for high risk and low risk disease and accurately guide the adjustment of subsequent treatment strategies.[13-15] In the current study, we investigated the role of the addition of PD-1 inhibitors in patients evaluated as having no radiological or biological response after induction chemotherapy, whom we defined as having high risk nasopharyngeal carcinoma. However, in those with radiological and biological response after induction chemotherapy, whom we defined as having low risk disease, we conducted another phase 3 trial (ClinicalTrials.com, NCT04448522) to explore the efficacy and safety profile of subsequent radiotherapy dose reduction strategies. These studies potentially validated the precision treatment concept of using dynamic imaging assessment and EBV DNA changes to individualise subsequent treatment intensity in nasopharyngeal carcinoma.

This phase 3 trial investigated the effect of PD-1 inhibitors in combination with concurrent chemoradiotherapy and as adjuvant treatment for high risk nasopharyngeal carcinoma. The study population included 40% of patients with stage N3 disease and 73% with stage 4a disease; both proportions were higher than in the CONTINNUMM[21] (33% and 70%, respectively) and DIPPER[30] (32% and 69%, respectively) studies. For the enrolled patients with stage 2-3 disease, all of whom were evaluated as having no radiological or biological response after induction chemotherapy, subsequent exploratory analyses showed that the progression-free survival rate was lower than that of patients with stage 4a disease evaluated as having a radiologicaland biological response. This finding suggested that immunotherapy should be added for patients who did not respond well to induction chemotherapy, even for stage 2 or T3N1 disease (previously considered to be low risk) to further improve outcomes in this population. Additionally, the three year progression-free survival rate of the control group in the current study was 71%, which was lower than the 76% reported in the CONTINNUMM[21] and DIPPER[30] studies for the same treatment.

Determining the most beneficial combination of immunotherapy with chemoradiotherapy remains a controversial issue. The DIPPER study showed that adjuvant PD-1 blockade significantly improved progression-free survival in patients with locoregionally advanced nasopharyngeal carcinoma, which translated into a 9.6% absolute enhancement of three year progression-free survival and a 44% reduction in the risk of disease relapse or death.[30] However, the current study recruited patients with higher risk disease and achieved a 12% absolute increase in the progression-free survival rate and a 50% reduction in the risk of disease relapse or death. These results might reflect the beneficial role of PD-1 inhibitor treatment concurrent with chemoradiotherapy and as maintenance treatment in nasopharyngeal carcinoma. Similarly, recent studies also showed that pembrolizumab plus concurrent chemoradiotherapy followed by pembrolizumab maintenance treatment improved overall survival and progression-free survival in patients with locally advanced cervical cancer.[31 32] While these findings support our study design, the specific contribution of each treatment phase to the overall efficacy and the optimal cycles of adjuvant maintenance treatment need further study.

The quality assurance committee at Sun Yat-sen University Cancer Center ensured quality standards were met when planning radiotherapy treatment for all patients in both groups. We found no significant differences in dosimetric parameters between the two groups. Among patients with locoregional relapse, an in-field relapse was observed in 13 of 14 patients (92.9%) in the camrelizumab group and 30 of 32 (93.8%) in the standard treatment group. Therefore, we believe that the combination of chemoradiotherapy and camrelizumab treatment increases progression-free survival and could be the standard of care for this group, mainly because of improved locoregional tumour control. However, there were similarities to a study by Chen and colleagues[33] when analysing progression-free survival in the camrelizumab group and the maintenance capecitabine group. Therefore, it remains to be seen if the added low dose maintenance chemotherapy can be regarded as a valid control in future nasopharyngeal carcinoma immunotherapy trials, and whether adjuvant capecitabine could yield the same efficacy results as camrelizumab alone.

Camrelizumab concurrent with chemoradiotherapy and as a maintenance treatment showed a favourable safety profile, with grade 3 or higher adverse events reported in 11% of patients. This rate is comparable to the DIPPER study, which suggested that concurrent PD-1 blockade did not increase the toxicity rate. This combination treatment further reduced the incidence of cherry hemangiomas from 89.7% in the DIPPER study[30] to 59.7% in the current study. Similarly, many studies have reported that the incidence of cherry hemangiomas has decreased, with reported grade 1-2 incidence of 21.7-57.9%, and grade 3 incidence of 0% when camrelizumab was combined with chemotherapy.[19 34 35] The biological mechanism involved could be that chemotherapy inhibits the hyperproliferation of skin capillary endothelial cells and this effect may be maintained for a long time, resulting in a decrease in the incidence of cherry hemangiomas.[19 35] Although 19 cycles of camrelizumab were added, quality of life scores were not significantly affected in patients in the camrelizumab group, especially in the financial difficulties category. Camrelizumab was provided free of charge to patients, and there were also commercial insurance reimbursements for some patients admitted to hospital owing to severe toxicity from camrelizumab. Additionally, most adverse reactions related to camrelizumab were grade 1-2, which mostly resolved after discontinuation and generally did not require special treatment. Importantly, camrelizumab decreased the risk of local recurrence and distant metastasis, which reduced the medical costs and economic burden from subsequent treatments for these patients.

Previous studies have shown that camrelizumab could be beneficial for patients with cervical cancer and head and neck cancer.[36-39] Concurrent chemoradiotherapy is the standard of care for patients with locally advanced unresectable cervical and head and neck cancer, although efficacy needs to be improved. Therefore, based on the findings of this study, we consider further clinical trials would be useful to analyse whether the combination of camrelizumab and conventional chemoradiotherapy could improve the outcomes of these patients.

### Limitations of this study

In spite of its encouraging findings, this study also has several limitations. Patients were enrolled from a region where nasopharyngeal carcinoma is predominantly linked to EBV infection, therefore the generalisability of our findings to other populations needs further validation. Additionally, this was an open label trial, which may have introduced biases that could be excluded in a double blind placebo controlled design. Nevertheless, several rigorous quality control measures were implemented throughout the study to minimise the likelihood of bias (supplementary appendix). Finally, all exploratory analyses should be interpreted with caution because they are not adjusted for multiplicity and generally represent preliminary findings.

### Conclusions

In conclusion, camrelizumab with concurrent chemoradiotherapy and as maintenance treatment significantly improved the progression-free survival rate with a favourable safety profile in patients with stage 4a nasopharyngeal carcinoma and in those with stage 2-3 disease evaluated to have no radiological or biological response after induction chemotherapy. Nevertheless, determining whether the 12% increase in progression-free survival at 36 months will translate into an overall survival benefit will require longer follow-up.


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**Ethical approval:** The study was approved by the institutional review board of the Sun Yat-sen University Cancer Center (B2020-131-01) and respective ethics committee of each participating centre.

**Data sharing:** The code used to analyse the data in the paper can be found in the supplemental files. The data underlying the findings in this paper are openly and publicly available in Dryad and can be found here: https://doi.org/10.5061/dryad.3j9kd51zg. If problems are encountered accessing the data, please contact the corresponding author.

**Transparency:** The lead author (M-YC) affirms that the manuscript is an honest, accurate, and transparent account of the study being reported; that no important aspects of the study have been omitted; and that any discrepancies from the study as planned have been explained."

**Dissemination to participants and related patient and public communities:** The findings of this study will be disseminated to participants, patient communities, and the general public through a press release. The press release will be distributed through our institutional website, official WeChat and LinkedIn accounts, and PR Newswire to ensure wide reaching visibility.

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1 Bray F, Laversanne M, Sung H, et al. Global cancer statistics 2022: GLOBOCAN estimates of incidence and mortality worldwide for 36 cancers in 185 countries. *CA Cancer J Clin* 2024;74:229-63. doi:10.3322/caac.21834.

2 NCCN Clinical Practice Guidelines in Oncology. *Head and Neck Cancers (Version 1.2020).* National Comprehensive Cancer Network, 2020.

3 Chen YP, Chan ATC, Le QT, Blanchard P, Sun Y, Ma J. Nasopharyngeal carcinoma. *Lancet* 2019;394:64-80. doi:10.1016/S0140-6736(19)30956-0.
4 Zhang Y, Chen L, Hu GQ, et al. Gemcitabine and cisplatin induction chemotherapy in nasopharyngeal carcinoma. *N Engl J Med* 2019;381:1124-35. doi:10.1056/NEJMoa1905287.
5 Sun Y, Li WF, Chen NY, et al. Induction chemotherapy plus concurrent chemoradiotherapy versus concurrent chemoradiotherapy alone in locoregionally advanced nasopharyngeal carcinoma: a phase 3, multicentre, randomised controlled trial. *Lancet Oncol* 2016;17:1509-20. doi:10.1016/S1470-2045(16)30410-7.
6 Yang Q, Cao SM, Guo L, et al. Induction chemotherapy followed by concurrent chemoradiotherapy versus concurrent chemoradiotherapy alone in locoregionally advanced nasopharyngeal carcinoma: long-term results of a phase III multicentre randomised controlled trial. *Eur J Cancer* 2019;119:87-96. doi:10.1016/j.ejca.2019.07.007.
7 Kurtz DM, Esfahani MS, Scherer F, et al. Dynamic risk profiling using serial tumor biomarkers for personalized outcome prediction. *Cell* 2019;178:699-713.e19. doi:10.1016/j.cell.2019.06.011.
8 Bedard PL, Hansen AR, Ratain MJ, Siu LL. Tumour heterogeneity in the clinic. *Nature* 2013;501:355-64. doi:10.1038/nature12627.
9 Barrington SF, Mikhaeel NG, Kostakoglu L, et al. Role of imaging in the staging and response assessment of lymphoma: consensus of the International Conference on Malignant Lymphomas Imaging Working Group. *J Clin Oncol* 2014;32:3048-58. doi:10.1200/JCO.2013.53.5229.
10 Cheson BD, Fisher RI, Barrington SF, et al, Alliance, Australasian Leukaemia and Lymphoma GroupEastern Cooperative Oncology GroupEuropean Mantle Cell Lymphoma ConsortiumItalian Lymphoma FoundationEuropean Organisation for ResearchTreatment of Cancer/Dutch Hemato-Oncology GroupGrupo Español de Médula ÓseaGerman High-Grade Lymphoma Study GroupGerman Hodgkin's Study GroupJapanese Lymphorra Study GroupLymphoma Study AssociationNCIC Clinical Trials GroupNordic Lymphoma Study GroupSouthwest Oncology GroupUnited Kingdom National Cancer Research Institute. Recommendations for initial evaluation, staging, and response assessment of Hodgkin and non-Hodgkin lymphoma: the Lugano classification. *J Clin Oncol* 2014;32:3059-68. doi:10.1200/JCO.2013.54.8800.
11 Hahnen E, Lederer B, Hauke J, et al. Germline mutation status, pathological complete response, and disease-free survival in triple-negative breast cancer: secondary analysis of the GeparSixto randomized clinical trial. *JAMA Oncol* 2017;3:1378-85. doi:10.1001/jamaoncol.2017.1007.
12 Ommen HB. Monitoring minimal residual disease in acute myeloid leukaemia: a review of the current evolving strategies. *Ther Adv Hematol* 2016;7:3-16. doi:10.1177/2040620715614529.
13 Peng H, Chen L, Zhang Y, et al. The tumour response to induction chemotherapy has prognostic value for long-term survival outcomes after intensity-modulated radiation therapy in nasopharyngeal carcinoma. *Sci Rep* 2016;6:24835. doi:10.1038/srep24835.
14 Lv J, Xu LX, Li ZX, et al. Longitudinal on-treatment circulating tumor DNA as a biomarker for real-time dynamic risk monitoring in cancer patients: The EP-SEASON study. *Cancer Cell* 2024;42:1401-1414.e4. doi:10.1016/j.ccell.2024.07.001.
15 Lv J, Chen Y, Zhou G, et al. Liquid biopsy tracking during sequential chemo-radiotherapy identifies distinct prognostic phenotypes in nasopharyngeal carcinoma. *Nat Commun* 2019;10:3941. doi:10.1038/s41467-019-11853-y.
16 Pan JJ, Mai HQ, Ng WT, et al. NinthvVersion of the AJCC and UICC nasopharyngeal cancer TNM staging classification. *JAMA Oncol* 2024;10:1627-35. doi:10.1001/jamaoncol.2024.4354.
17 Du XJ, Wang GY, Zhu XD, et al. Refining the 8th edition TNM classification for EBV related nasopharyngeal carcinoma. *Cancer Cell* 2024;42:464-473.e3. doi:10.1016/j.ccell.2023.12.020.
18 Mai HQ, Chen QY, Chen D, et al. Toripalimab plus chemotherapy for recurrent or metastatic nasopharyngeal carcinoma: the JUPITER-02 randomized clinical trial. *JAMA* 2023;330:1961-70. doi:10.1001/jama.2023.20181.
19 Yang Y, Qu S, Li J, et al. Camrelizumab versus placebo in combination with gemcitabine and cisplatin as first-line treatment for recurrent or metastatic nasopharyngeal carcinoma (CAPTAIN-1st): a multicentre, randomised, double-blind, phase 3 trial. *Lancet Oncol* 2021;22:1162-74. doi:10.1016/S1470-2045(21)00302-8.
20 Yang Y, Pan J, Wang H, et al. Tislelizumab plus chemotherapy as first-line treatment for recurrent or metastatic nasopharyngeal cancer: a multicenter phase 3 trial (RATIONALE-309). *Cancer Cell* 2023;41:1061-1072.e4. doi:10.1016/j.ccell.2023.04.014.
21 Liu X, Zhang Y, Yang KY, et al. Induction-concurrent chemoradiotherapy with or without sintilimab in patients with locoregionally advanced nasopharyngeal carcinoma in China (CONTINUUM): a multicentre, open-label, parallel-group, randomised, controlled, phase 3 trial. *Lancet* 2024;403:2720-31. doi:10.1016/S0140-6736(24)00594-4.
22 Ding X, Zhang WJ, You R, et al. Camrelizumab plus apatinib in patients with recurrent or metastatic nasopharyngeal carcinoma: an open-label, single-arm, phase II study. *J Clin Oncol* 2023;41:2571-82. doi:10.1200/JCO.22.01450.
23 Guo SS, Tang LQ, Zhang L, et al. The impact of the cumulative dose of cisplatin during concurrent chemoradiotherapy on the clinical outcomes of patients with advanced-stage nasopharyngeal carcinoma in an era of intensity-modulated radiotherapy. *BMC Cancer* 2015;15:977. doi:10.1186/s12885-015-1964-8.
24 Lee AW, Tung SY, Ngan RK, et al. Factors contributing to the efficacy of concurrent-adjuvant chemotherapy for locoregionally advanced nasopharyngeal carcinoma: combined analyses of NPC-9901 and NPC-9902 trials. *Eur J Cancer* 2011;47:656-66. doi:10.1016/j.ejca.2010.10.026.
25 Ng WT, Tung SY, Lee V, et al. Concurrent-adjuvant chemoradiation therapy for stage III-IVB nasopharyngeal carcinoma-exploration for achieving optimal 10-year therapeutic ratio. *Int J Radiat Oncol Biol Phys* 2018;101:1078-86. doi:10.1016/j.ijrobp.2018.04.069.
26 Peng H, Chen L, Zhang Y, et al. Prognostic value of the cumulative cisplatin dose during concurrent chemoradiotherapy in locoregionally advanced nasopharyngeal carcinoma: a secondary analysis of a prospective phase III clinical trial. *Oncologist* 2016;21:1369-76. doi:10.1634/theoncologist.2016-0105.
27 Cengiz M, Ozyar E, Esassolak M, et al. Assessment of quality of life of nasopharyngeal carcinoma patients with EORTC QLQ-C30 and H&N-35 modules. *Int J Radiat Oncol Biol Phys* 2005;63:1347-53. doi:10.1016/j.ijrobp.2005.05.057.
28 Schoenfeld DA. Sample-size formula for the proportional-hazards regression model. *Biometrics* 1983;39:499-503. doi:10.2307/2531021
29 Dawson LA, Anzai Y, Marsh L, et al. Patterns of local-regional recurrence following parotid-sparing conformal and segmental intensity-modulated radiotherapy for head and neck cancer. *Int J Radiat Oncol Biol Phys* 2000;46:1117-26. doi:10.1016/S0360-3016(99)00550-7.
30 Liang YL, Liu X, Shen LF, et al. Adjuvant PD-1 blockade with camrelizumab for nasopharyngeal carcinoma: the DIPPER randomized clinical trial. *JAMA* 2025;333:1589-98. doi:10.1001/jama.2025.1132.
31 Lorusso D, Xiang Y, Hasegawa K, et al, ENGOT-cx11/GOG-3047/KEYNOTE-A18 investigators. Pembrolizumab or placebo with chemoradiotherapy followed by pembrolizumab or placebo for newly diagnosed, high-risk, locally advanced cervical cancer (ENGOT-cx11/GOG-3047/KEYNOTE-A18): a randomised, double-blind, phase 3 clinical trial. *Lancet* 2024;403:1341-50. doi:10.1016/S0140-6736(24)00317-9.
32 Lorusso D, Xiang Y, Hasegawa K, et al, ENGOT-cx11/GOG-3047/KEYNOTE-A18 investigators. Pembrolizumab or placebo with chemoradiotherapy followed by pembrolizumab or placebo for newly diagnosed, high-risk, locally advanced cervical cancer (ENGOT-cx11/GOG-3047/KEYNOTE-A18): overall survival results from a randomised, double-blind, placebo-controlled, phase 3 trial. *Lancet* 2024;404:1321-32. doi:10.1016/S0140-6736(24)01808-7.
33 Chen YP, Liu X, Zhou Q, et al. Metronomic capecitabine as adjuvant therapy in locoregionally advanced nasopharyngeal carcinoma: a multicentre, open-label, parallel-group, randomised, controlled, phase 3 trial. *Lancet* 2021;398:303-13. doi:10.1016/S0140-6736(21)01123-5.
34 Wang Z, Sun Y, Wang Q, et al. Induction chemotherapy plus camrelizumab combined with concurrent chemoradiotherapy in locoregionally advanced nasopharyngeal carcinoma in non-endemic areas: a phase 2 clinical trial in North China. *BMC Med* 2025;23:126. doi:10.1186/s12916-025-03964-9.
35 Fang W, Yang Y, Ma Y, et al. Camrelizumab (SHR-1210) alone or in combination with gemcitabine plus cisplatin for nasopharyngeal carcinoma: results from two single-arm, phase 1 trials. *Lancet Oncol* 2018;19:1338-50. doi:10.1016/S1470-2045(18)30495-9.
36 Xiang Z, Wei X, Zhang Z, et al. Efficacy, safety and single-cell analysis of neoadjuvant immunochemotherapy in locally advanced oral squamous cell carcinoma: a phase II trial. *Nat Commun* 2025;16:3968. doi:10.1038/s41467-025-59004-w.
37 Liu HM, Xiong XP, Yu ZL, et al. Neoadjuvant immunotherapy with or without chemotherapy in locally advanced oral squamous cell carcinoma: Randomized, two-arm, phase 2 trial. *Cell Rep Med* 2025;6:101930. doi:10.1016/j.xcrm.2025.101930.
38 Xia L, Zhang K, Tang Y, et al. Camrelizumab plus famitinib versus camrelizumab alone and investigator's choice of chemotherapy in recurrent or metastatic cervical cancer: a randomized, phase II study. *J Clin Oncol* 2025;43:2720-33. doi:10.1200/JCO-24-02495.
39 Mi X, Tuo F, Lin T. Efficacy and safety of camrelizumab for the treatment of cervical cancer: a systematic review and meta-analysis. *Front Oncol* 2024;14:1526103. doi:10.3389/fonc.2024.1526103.

**Web appendix:** Supplementary appendix